

On Robustness of Adaptive Feedback Control for Stock Trading With Time-Varying Price Dynamics

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Abstract—We present an adaptive form of the simultaneous long-short (SLS) trading strategy, which employs feedback control principles for stock trading. Prior work established the “robust positive expectation” (RPE) theorem for the SLS strategy, asserting positive expectations of trading profits for stock prices following geometric Brownian motions with unknown parameters. We extend the RPE property to adaptive SLS strategies, which guarantee strictly positive expected profits even in round-trip price scenarios where traditional SLS strategies fail.

Index Terms—Financial engineering, robust adaptive control, stochastic systems, time-varying systems.

I. INTRODUCTION

A New line of research, for example, see [1], [2], [3], [4], [5], [6], and [7], has explored a model-free and reactive approach to stock trading based on principles from feedback control. Known as the simultaneous long-short (SLS) trading strategy [2], this approach uses a combination of long and short linear feedback controllers to determine investment levels, without relying on predictive modeling of stock price dynamics. A key finding from the seminal works, see [3] for a unified description, is that under idealized market conditions where stock prices follow a geometric Brownian motion (GBM), the SLS strategy can achieve a positive expected trading gain, a property termed the “robust positive expectation (RPE).” This property has been shown to be independent of the underlying parameters of the GBM.

In line with the model-free control-theoretical approach, recent studies have revealed numerous extensions and significant generalizations. For instance, the RPE theorem of the SLS has been extended to Merton’s jump diffusion model [4] and time-varying GBM [8]. Other works propose generalizations of the original proportional SLS controller to the initially long-short (ILS) feedback controller with delay [9], proportional–integral SLS [5], and SLS with asymmetric feedback gains for long and short positions [6]. These contributions have significantly enriched the family of SLS controllers and strengthened the control-theoretic approach to stock trading, where the focus is on ensuring an RPE of profits. Another critical topic is the selection of SLS controller parameters for enhanced trading performance, particularly in a time-varying market where tuning the parameters poses a considerable challenge. While proposals have been made to adapt the controller to time-varying price dynamics, such as replacing the fixed feedback gain

with an extremum seeking controller to obtain dynamically optimized gains [10], and applying a variable moving window to the most recent price time series to search for optimal gains [11], the RPE property of adaptive SLS strategies has not been explicitly proven.

Recognizing the central role of the RPE in the control-theoretical approach to stock trading and the necessity of adaptive control for realistic, time-varying markets, this work aims to establish the RPE theorem for adaptive SLS strategies trading against time-varying price dynamics. The resulting theorem demonstrates that the fixed parameters of the SLS controller, namely, the initial investment level, I_0 , and the feedback gain, K , can be replaced by time-varying functions while still maintaining the RPE property. The time-varying counterparts of I_0 and K , denoted as $\tilde{I}(t)$ and $\tilde{K}(t)$, respectively, can be arbitrary differentiable positive functions as long as the ratio $\frac{\tilde{I}(t)}{\tilde{K}(t)}$ is nondecreasing. Furthermore, we find that the previous condition for strictly positive profit [8], represented as

$$\int_0^t \mu(\tau) d\tau \neq 0$$

where μ is the drift parameter of GMB, can be relaxed to

$$\mu(\tau) \neq 0 \quad \text{on } [0, t]$$

if $\frac{\tilde{I}(t)}{\tilde{K}(t)}$ is strictly increasing. In other words, adaptive SLS strategies can achieve strictly positive expected gains as long as the drift function is not identically zero over the entire trading period, even if the time-averaged drift is zero. This theorem represents a significant step forward in the class of SLS controllers for stock trading, as it achieves the seemingly “impossible”: it guarantees a strictly positive expected gain for a market exhibiting uptrends, downtrends, and oscillating trends in a purely reactive manner.

The rest of this article is organized as follows. Section II reviews the original SLS strategy and its RPE property. Section III derives the expected gain of adaptive SLS strategies. Section IV establishes the robustness theorem of adaptive SLS strategies and provides an illustrative example. Section V presents the simulation results with real-world financial data. Finally, Section VI concludes this article.

II. SLS STRATEGY AND RPE PROPERTY

The formulation of the stock-trading process within the context of a classical feedback control setting was initially proposed by Barmish [1], and later developed further in collaboration with Primbs [2]. It is worth noting that Dokuchaev and Savkin [12], [13] had earlier studied similar trading strategies with positive expected gains without explicitly mentioning feedback control principles. While their work provided valuable insights into the positive expectation property, this section presents the SLS and its RPE property from a control-theoretic perspective, as introduced by Barmish and Primbs [3]. The basic assumption is the idealized frictionless market as follows:

Received 30 July 2024; revised 14 November 2024; accepted 16 November 2024. Date of publication 19 November 2024; date of current version 28 April 2025. This work was supported by the National Natural Science Foundation of China under Grant 72103102. Recommended by Associate Editor M. E. Broucke. (Corresponding author: Feng Zhu.)

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Digital Object Identifier 10.1109/TAC.2024.3502506

- 1) absence of transaction costs (e.g., commissions, fees, slippage, interest, taxes, and bid-ask spread);
- 2) continuous trading of unlimited and fractional shares;
- 3) instantaneous execution of trades without delay;
- 4) no price impact from trading actions.

Consider a stock price $p(t)$ governed by a GBM

$$\frac{dp}{p} = \mu dt + \sigma dW(t)$$

where μ is the drift rate, σ is the volatility (annualized standard deviation), and $W(t)$ is a Wiener process.

Let $I(t)$ denote the investment level, a control input to the trading account, which can be positive or negative. A positive $I(t)$ indicates that the trader is taking a long position, profiting from a price increase, while a negative $I(t)$ indicates a short position, profiting from a price decrease. Let $g(t)$ represent the cumulative gain from the start of the trading horizon up to time t . The SLS strategy involves maintaining two accounts. The long account, with investment levels $I_L(t)$, is determined by its cumulative gain $g_L(t)$ through the linear feedback principle

$$I_L(t) = I_0 + K g_L(t) \quad (1)$$

where $I_0 > 0$ is the initial investment level, and $K > 0$ is the feedback gain. The short account has the same initial investment level and feedback gain; however, its investment level is determined by its own cumulative gain, $g_S(t)$, also via the linear feedback principle

$$I_S(t) = -I_0 - K g_S(t). \quad (2)$$

During an infinitesimally small time interval $[t, t + dt]$, we have the following incremental relations:

$$dg_L(t) = \frac{dp}{p} I_L(t) = (\mu dt + \sigma dW(t)) (I_0 + K g_L(t))$$

$$dg_S(t) = \frac{dp}{p} I_S(t) = -(\mu dt + \sigma dW(t)) (I_0 + K g_S(t))$$

where the first equality in each equation can be interpreted as “the difference in the cumulative gain is equal to the difference in price times the number of shares (i.e., $\frac{I}{p}$) during the infinitesimally small time interval.” The above closed-loop trading systems can be solved to yield the following closed-form solutions (proof is in [3])

$$g_L(t) = \frac{I_0}{K} \left(\left(\frac{p(t)}{p(0)} \right)^K e^{\frac{1}{2}\sigma^2(K-K^2)t} - 1 \right)$$

$$g_S(t) = \frac{I_0}{K} \left(\left(\frac{p(t)}{p(0)} \right)^{-K} e^{-\frac{1}{2}\sigma^2(K+K^2)t} - 1 \right).$$

The expected gain of the SLS strategy is (proof is in [3])

$$\mathbb{E}[g(t)] := \mathbb{E}[g_L(t) + g_S(t)] = \frac{I_0}{K} (e^{\mu K t} + e^{-\mu K t} - 2). \quad (3)$$

It is seen that the expected gain is positive for all time $t > 0$ if $\mu \neq 0$, as $e^x + e^{-x} > 2$ if $x \neq 0$. This is the “robust positive expectation” property of the original form of SLS strategy.

III. EXPECTED GAIN OF ADAPTIVE SLS STRATEGIES

In this section, we consider the SLS strategy trading against continuously time-varying GBM

$$\frac{dp}{p} = \mu(t)dt + \sigma(t)dW(t) \quad (4)$$

where $\mu(t)$ and $\sigma(t)$ are time-varying drift and volatility, respectively. The time-varying GBM is a more realistic model of the price process than the GBM.

We enable the feedback gain to be a positive differentiable function, $K(t)$, with $K(0) = K_0$. Furthermore, the investment level is directly adjusted via a positive differentiable function $\tilde{I}(t)$, starting with an initial value of $\tilde{I}(0) = I_0$. Consequently, the investment levels for long and short positions adapt as follows:

$$I_L(t) = \tilde{I}(t) + K(t)g_L(t) \quad (5)$$

$$I_S(t) = -\tilde{I}(t) - K(t)g_S(t). \quad (6)$$

These adaptive investment levels generalize the original SLS strategy by replacing the constant initial investment I_0 and feedback gain K in (1) and (2) with their time-varying counterparts $\tilde{I}(t)$ and $K(t)$, respectively.

The closed-loop stochastic trading systems for long and short positions are as follows:

$$dg_L = (\mu(t)dt + \sigma(t)dW(t)) (\tilde{I}(t) + K(t)g_L) \quad (7)$$

$$dg_S = (\mu(t)dt + \sigma(t)dW(t)) (-\tilde{I}(t) - K(t)g_S). \quad (8)$$

The above equations are linear and have closed-form solutions and expectations of solutions according to Theorem 1.

Theorem 1 (See [14, Ch. 9]): Consider the standard form of the linear stochastic differential equation as follows:

$$dX(t) = (\alpha(t)X(t) + \beta(t))dt + (\varphi(t)X(t) + \vartheta(t))dW(t)$$

where $\alpha(t)$, $\beta(t)$, $\varphi(t)$, and $\vartheta(t)$ are real-valued nonrandom functions bounded on every finite interval, and $X(0) = x_0$. The linear stochastic differential equation has a unique closed-form solution. The expectation of the solution is

$$\mathbb{E}[X(t)] = e^{\int_0^t \alpha(v)dv} \left(x_0 + \int_0^t \beta(s)e^{-\int_0^s \alpha(\eta)d\eta} ds \right).$$

Applying Theorem 1 to the adaptive SLS, we have the following lemma.

Lemma 1: The expectation of the gain-loss function of the closed-loop trading systems defined by (7) and (8) is

$$\mathbb{E}[g(t)] = 2 \int_0^t \mu(s)\tilde{I}(s) \sinh \left(\int_s^t \mu(\eta)K(\eta)d\eta \right) ds \quad (9)$$

which is independent of the volatility function, σ .

Proof: Using Theorem 1, the expectation of the gain-loss function is

$$\begin{aligned} \mathbb{E}[g(t)] &= \mathbb{E}[g_L(t)] + \mathbb{E}[g_S(t)] \\ &= e^{\int_0^t \mu(v)K(v)dv} \int_0^t \mu(s)\tilde{I}(s)e^{-\int_0^s \mu(\eta)K(\eta)d\eta} ds \\ &\quad - e^{-\int_0^t \mu(v)K(v)dv} \int_0^t \mu(s)\tilde{I}(s)e^{\int_0^s \mu(\eta)K(\eta)d\eta} ds \\ &= \int_0^t \mu(s)\tilde{I}(s) \left(\frac{e^{\int_0^t \mu(v)K(v)dv}}{e^{\int_0^s \mu(\eta)K(\eta)d\eta}} - \frac{e^{\int_0^s \mu(\eta)K(\eta)d\eta}}{e^{\int_0^t \mu(v)K(v)dv}} \right) ds \\ &= 2 \int_0^t \mu(s)\tilde{I}(s) \sinh \left(\int_s^t \mu(\eta)K(\eta)d\eta \right) ds \end{aligned}$$

where σ is not involved in the expected-value formula. ■

Remark 1: A special case of Lemma 1 is that when $\mu(t)$, $\sigma(t)$, $K(t)$, and $\tilde{I}(t)$ are all constants with the values of μ , σ , K , and I_0 ,

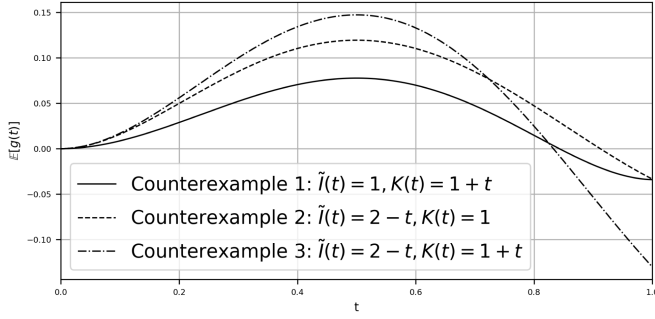


Fig. 1. Expected gains are calculated by (9) with $\mu(t) = 1 - 2t$, demonstrating the failure of the RPE property when the market trend is not unidirectional.

respectively, then

$$\begin{aligned}\mathbb{E}[g(t)] &= 2\mu I_0 \int_0^t \sinh((t-s)K\mu) ds \\ &= 2\frac{I_0}{K} (\cosh(tK\mu) - 1) = \frac{I_0}{K} (e^{\mu K t} + e^{-\mu K t} - 2)\end{aligned}$$

which is (3) such that $\mathbb{E}[g(t)] > 0$ for all $t > 0$ if $\mu \neq 0$.

Remark 2: Another special case of Lemma 1 is that when $K(t) = K$ and $\tilde{I}(t) = I_0$ are constant functions, the drifts $\mu(t)$ and $\sigma(t)$ are time-varying. Let $M(t) = \int_0^t \mu(\eta) d\eta$. Then

$$\begin{aligned}\mathbb{E}[g(t)] &= 2 \int_0^t \mu(s) I_0 \sinh\left(\int_s^t \mu(\eta) K d\eta\right) ds \\ &= 2I_0 \int_0^t \sinh(KM(t) - KM(s)) dM(s) \\ &= -\frac{2I_0}{K} \cosh(K(M(t) - M(s))) \Big|_0^t \\ &= \frac{2I_0}{K} (\cosh(KM(t)) - 1) \\ &= \frac{2I_0}{K} \left(\cosh\left(K \int_0^t \mu(\eta) d\eta\right) - 1 \right).\end{aligned}\quad (10)$$

It is seen that $\mathbb{E}[g(t)] > 0$ for all $t > 0$ if $\int_0^t \mu(\eta) d\eta \neq 0$. An equivalent result of (10) is first derived by Dokuchaev and Savkin [12], [13], considering both stocks following GBM and a risk-free bond. Setting aside the bond component yields the result of (10), which is also arrived at independently by Primbs and Barmish [8] from a control-theoretic perspective. The result of (10) is significant, as it extends the RPE property of the SLS strategy to time-varying price dynamics.

Remark 3: It is generally not true that the SLS controllers have the RPE property when their parameters are time-varying under a time-varying GBM, as seen in Fig. 1. However, if the drift $\mu(t)$ does not change its sign during the trading period of $[0, T]$ (i.e., $\mu(t) > 0$ for all $t \in [0, T]$ or $\mu(t) < 0$ for all $t \in [0, T]$), then $\mathbb{E}[g(t)] > 0$ for all $t \in (0, T]$. This is because $\sinh(\cdot)$ is an odd function, so the μ in the outer integral and the nested-integral together will produce a nonnegative sign in (9).

IV. ROBUSTNESS OF ADAPTIVE SLS STRATEGIES

In real-world scenarios, it is often unrealistic to assume that market trends will remain unidirectional. Failing to adjust the parameters of the SLS controller when a trend reversal occurs can result in the loss of profits gained during the initial trend. However, no existing theorem lays

out the rules for adapting the controller parameters without compromising the RPE property. Our objective is to determine the conditions under which the functions K and/or $\tilde{I}$ can be adjusted without imposing constraints on the drift function μ while still maintaining the RPE property. The following theorem is our key finding on such a sufficient condition.

Theorem 2: Consider the expected trading gain $\mathbb{E}[g(t)]$ given by (9) for the closed-loop trading systems defined by (7) and (8). Suppose $\tilde{I}$ and K are positive differentiable functions on $[0, t]$, and let $u(s) := \frac{\tilde{I}(s)}{K(s)}$. Then the following holds.

- 1) If $u'(s) \geq 0$ for all $s \in [0, t]$, then $\mathbb{E}[g(t)] \geq 0$.
- 2) If $u'(s) > 0$ for all $s \in [0, t]$ and the drift function $\mu(s) \neq 0$ on $[0, t]$, then $\mathbb{E}[g(t)] > 0$.

Proof: Let $v(s) := \cosh\left(\int_s^t \mu(\eta) K(\eta) d\eta\right)$, then $\frac{dv}{ds} = -\mu(s)K(s) \sinh\left(\int_s^t \mu(\eta) K(\eta) d\eta\right)$. Substituting $u(s)$ and $\frac{dv}{ds}$ into (9), we obtain

$$\mathbb{E}[g(t)] = -2 \int_0^t u(s) \frac{dv}{ds} ds.$$

Using integration by parts, we have

$$\mathbb{E}[g(t)] = -2 \left[u(s)v(s) \Big|_{s=0}^{s=t} - \int_0^t \frac{du}{ds} v(s) ds \right].$$

Evaluating the boundary terms gives

$$u(t)v(t) = u(t)$$

$$u(0)v(0) = u(0) \cosh\left(\int_0^t \mu(\eta) K(\eta) d\eta\right).$$

Hence

$$\begin{aligned}\mathbb{E}[g(t)] &= 2 \left[u(0) \cosh\left(\int_0^t \mu(\eta) K(\eta) d\eta\right) - u(t) \right. \\ &\quad \left. + \int_0^t \frac{du}{ds} \cosh\left(\int_s^t \mu(\eta) K(\eta) d\eta\right) ds \right].\end{aligned}\quad (11)$$

Since $\frac{du}{ds} \geq 0$, and $\cosh(\cdot) \geq 1$, we have

$$\begin{aligned}\int_0^t \frac{du}{ds} \cosh\left(\int_s^t \mu(\eta) K(\eta) d\eta\right) ds \\ \geq \int_0^t \frac{du}{ds} \cdot 1 ds = u(t) - u(0).\end{aligned}\quad (12)$$

Substituting (12) in (11), we obtain

$$\mathbb{E}[g(t)] \geq 2u(0) \left[\cosh\left(\int_0^t \mu(\eta) K(\eta) d\eta\right) - 1 \right] \geq 0.$$

Moreover, if $\mu(s)$ is not identically zero and $\frac{du}{ds} > 0$ on $[0, t]$, then $\mathbb{E}[g(t)] > 0$. To show that this is the case, we just need to show the inequality in (12) is strict.

First note that function $\int_s^t \mu(\eta) K(\eta) d\eta$ is a continuous function of s . We claim that it must be nonzero for some s in $[0, t]$; otherwise, we would have $\mu(\eta)K(\eta) = 0$ almost everywhere. Since $K(\eta)$ is positive, this would imply that the drift is zero almost everywhere, which contradicts our assumption on μ . For those s in $[0, t]$ such that $\int_s^t \mu(\eta) K(\eta) d\eta \neq 0$, $\cosh\left(\int_s^t \mu(\eta) K(\eta) d\eta\right) > 1$. Since $\frac{du}{ds} > 0$ and $\cosh\left(\int_s^t \mu(\eta) K(\eta) d\eta\right) > 1$ for some s in $[0, t]$, the inequality in (12) is strict. ■

Remark 4: Theorem 2 establishes that the adaptive SLS controller preserves the RPE property. The theorem's significance lies in its relaxation of the condition for a strictly positive expectation of profit.

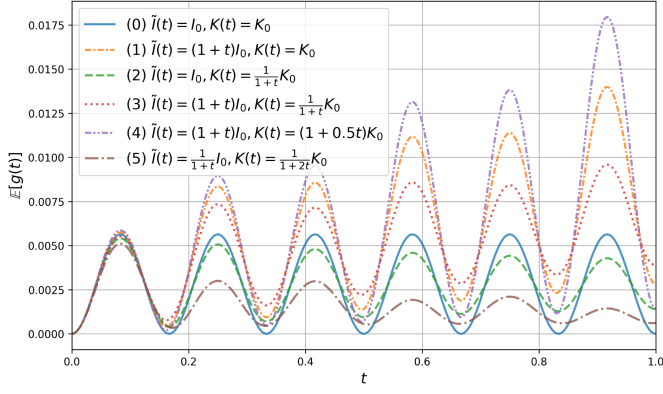


Fig. 2. Expected gains are calculated by (9) with $\mu(t) = \cos 6\pi t$. In all cases [except for the base case (0)], $\mathbb{E}[g(t)] > 0$ on $(0, t]$.

While Remark 2 requires $\int_0^t \mu(u) du \neq 0$, Theorem 2 only necessitates $\mu(s) \neq 0$ on $[0, t]$. In other words, the adaptive SLS strategy can yield a strictly positive expected gain as long as the drift function is not the zero function, even if the time-averaged drift is zero.

Remark 5: The flexibility in choosing $\tilde{I}$ and K functions allows for diverse adaptation strategies. For instance, the following choices of $\tilde{I}$ and K will satisfy the following condition.

- 1) $\tilde{I}$ is increasing while K remains constant.
- 2) K is decreasing while $\tilde{I}$ remains constant.
- 3) $\tilde{I}$ is increasing while K is decreasing.
- 4) Both $\tilde{I}$ and K are increasing, but $\tilde{I}$ grows faster than K .
- 5) Both $\tilde{I}$ and K are decreasing, but K decreases faster than $\tilde{I}$.

In addition, combinations of the above cases applied piecewise are also acceptable to satisfy the RPE property. Example 1 illustrates each of the above cases, along with a base case where K and $\tilde{I}$ are constant functions.

Example 1: Consider an adaptive SLS controller with initial parameters $I_0 = 1$ and $K_0 = 2$. We examine the following combinations of $\tilde{I}(t)$ and $K(t)$, as shown in Fig. 2.

- 0) $\tilde{I}(t) = I_0$ and $K(t) = K_0$: Both $\tilde{I}(t)$ and $K(t)$ are constant functions.
- 1) $\tilde{I}(t) = (1+t)I_0$ and $K(t) = K_0$: $\tilde{I}(t)$ increases linearly with time while $K(t)$ remains constant.
- 2) $\tilde{I}(t) = I_0$ and $K(t) = \frac{1}{1+t}K_0$: $\tilde{I}(t)$ remains constant while $K(t)$ decreases as a reciprocal function of time.
- 3) $\tilde{I}(t) = (1+t)I_0$ and $K(t) = \frac{1}{1+t}K_0$: $\tilde{I}(t)$ increases linearly with time while $K(t)$ decreases as a reciprocal function of time.
- 4) $\tilde{I}(t) = (1+t)I_0$ and $K(t) = (1+0.5t)K_0$: Both $\tilde{I}(t)$ and $K(t)$ increase linearly with time, but $\tilde{I}(t)$ grows twice as fast as $K(t)$.
- 5) $\tilde{I}(t) = \frac{1}{1+t}I_0$ and $K(t) = \frac{1}{1+2t}K_0$: Both $\tilde{I}(t)$ and $K(t)$ decrease as reciprocal functions of time, but $K(t)$ decreases twice as fast as $\tilde{I}(t)$.

V. SIMULATION OF AN ADAPTIVE SLS STRATEGY TRADING WITH REAL-WORLD FINANCIAL DATA

This section presents an empirical backtesting of an adaptive SLS strategy using NASDAQ-100 stocks and compares its performance with the original SLS strategy. We implement the adaptive SLS strategy from case (3) of Example 1, which exhibits a balance between maximizing expected gain and stabilizing fluctuations. The simulation uses daily stock prices from 25 September 2013 to 25 September 2023, obtained from yfinance [15]. To facilitate discrete-time trading, we employ the following update equations to approximate

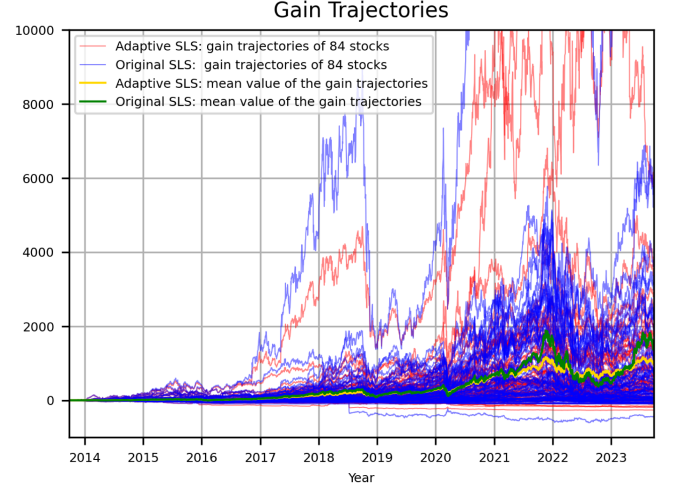


Fig. 3. Gain trajectories of the original SLS and the adaptive SLS strategies.

the continuous control policy: $I_L(k) = (1 + \frac{k}{N})I_0 + \frac{1}{1+k/N}K_0g_L(k)$, $I_S(k) = -(1 + \frac{k}{N})I_0 - \frac{1}{1+k/N}K_0g_S(k)$, $g_L(k+1) = g_L(k) + \frac{p(k+1)-p(k)}{p(k)}I_L(k)$, and $g_S(k+1) = g_S(k) + \frac{p(k+1)-p(k)}{p(k)}I_S(k)$, where $N = 2517$ is the number of trading days, and $I_0 = 1$ and $K_0 = 2$. The original SLS strategy uses the same update rule without the k/N term, as employed in [3]. We note that this discretization approach warrants further formal investigation, beyond this empirical backtesting, to determine its implications for different trading frequencies.

Due to data availability constraints, the simulation uses a subset of eighty-six NASDAQ-100 stocks with complete data for the specified period. The stock tickers are: AAPL, ADI, ADP, ADSK, AEP, ALGN, AMAT, AMD, AMGN, AMZN, ANSS, ASML, ATVI, AVGO, AZN, BIIB, BKNG, BKR, CDNS, CHTR, CMCSA, COST, CPRT, CSCO, CSGP, CSX, CTAS, CTSH, DLTR, DXCM, EA, EBAY, ENPH, EXC, FANG, FAST, FI, FTNT, GILD, GOOG, GOOGL, HON, IDXX, ILMN, INTC, INTU, ISRG, KDP, KLAC, LRCX, LULU, MAR, MCHP, MDLZ, MELI, META, MNST, MRVL, MSFT, MU, NFLX, NVDA, NXPI, ODFL, ON, ORLY, PANW, PAYX, PCAR, PEP, QCOM, REGN, ROST, SBUX, SGEN, SIRI, SNPS, TMUS, TSLA, TXN, VRSK, VRTX, WBA, WBD, WDAY, and XEL.

Fig. 3 presents the gain trajectories for the 86 stocks over a ten-year trading horizon, along with the mean values of the trajectories for both the original SLS and adaptive SLS strategies. The results show that at the end of the trading horizon, most stocks exhibit positive gains for both strategies, and the original SLS has a higher average gain. However, it is important to note that the original SLS and the adaptive SLS have different investment levels throughout the trading period, which makes a direct comparison of their gains less meaningful.

For a more meaningful comparison, we introduce the return-on-maximum-investment (ROI) metric, defined as $ROI(t) = \frac{g(t)}{\max_{\tau \in [0, t]} |I(\tau)|}$, which accounts for different investment levels. Fig. 4 presents the mean ROIs and percentage of positive gains for both strategies. The adaptive SLS achieves significantly higher ROIs over time, indicating better return efficiency relative to maximum investment. However, it also shows higher variance, suggesting greater performance variability across stocks. The percentage of positive gains also increases over time, with the adaptive SLS gradually outperforming

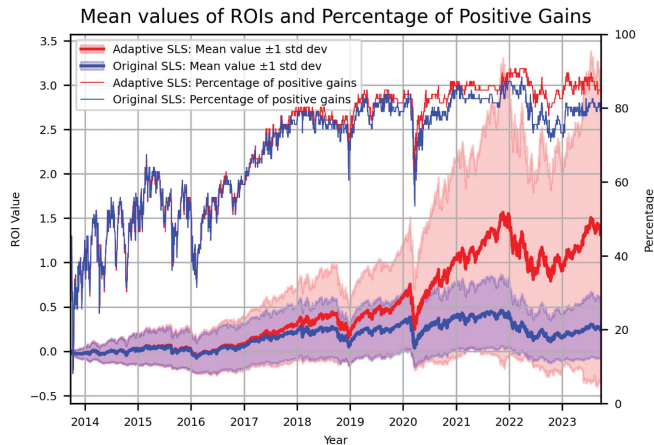


Fig. 4. Mean ROI and percentage of positive gains of the original SLS and the adaptive SLS strategies.

the original SLS, aligning with less restrictive conditions for positive gains as per our robustness theorem.

VI. CONCLUSION

In this work, we have extended the robust positive expectation property of the SLS trading strategy to adaptive control settings under time-varying market conditions. Our main contribution is the establishment of the RPE theorem for adaptive SLS strategies, demonstrating that the fixed parameters of the original SLS controller can be replaced by time-varying functions while maintaining the RPE property. Specifically, we showed that the initial investment level and feedback gain can be adapted as long as their ratio is nondecreasing over time. This adaptive approach not only preserves the RPE property but also relaxes the conditions required for strictly positive expected gains, allowing the SLS strategy to achieve profits even under oscillating market trends.

Future work might focus on several key areas. First, analyzing discrete-time formulations of adaptive SLS strategies could yield more actionable insights for real-world trading applications. Second, employing control techniques to dynamically calibrate the parameters of the adaptive SLS based on real-time market conditions could potentially enhance its profitability. In addition, incorporating transaction costs and market impact into the SLS framework would provide a more comprehensive understanding of its real-world applicability.

ACKNOWLEDGMENT

The authors would like to thank the anonymous reviewers whose insightful comments significantly enhanced the technical quality of the

manuscript. Initially, our theorem 2 focused on scaling the parameters I_0 and K by a common factor. One of the reviewers suggested considering an investment level function with different $\tilde{I}(t)$ and $K(t)$, which led to a revised theorem 2 involving a polynomial function of $\tilde{I}$ with nonnegative coefficients in addition to the common factor. Another reviewer further generalized our theorem 2 to accommodate nondecreasing $\tilde{I}$, and subsequently extended it to its current setup.

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